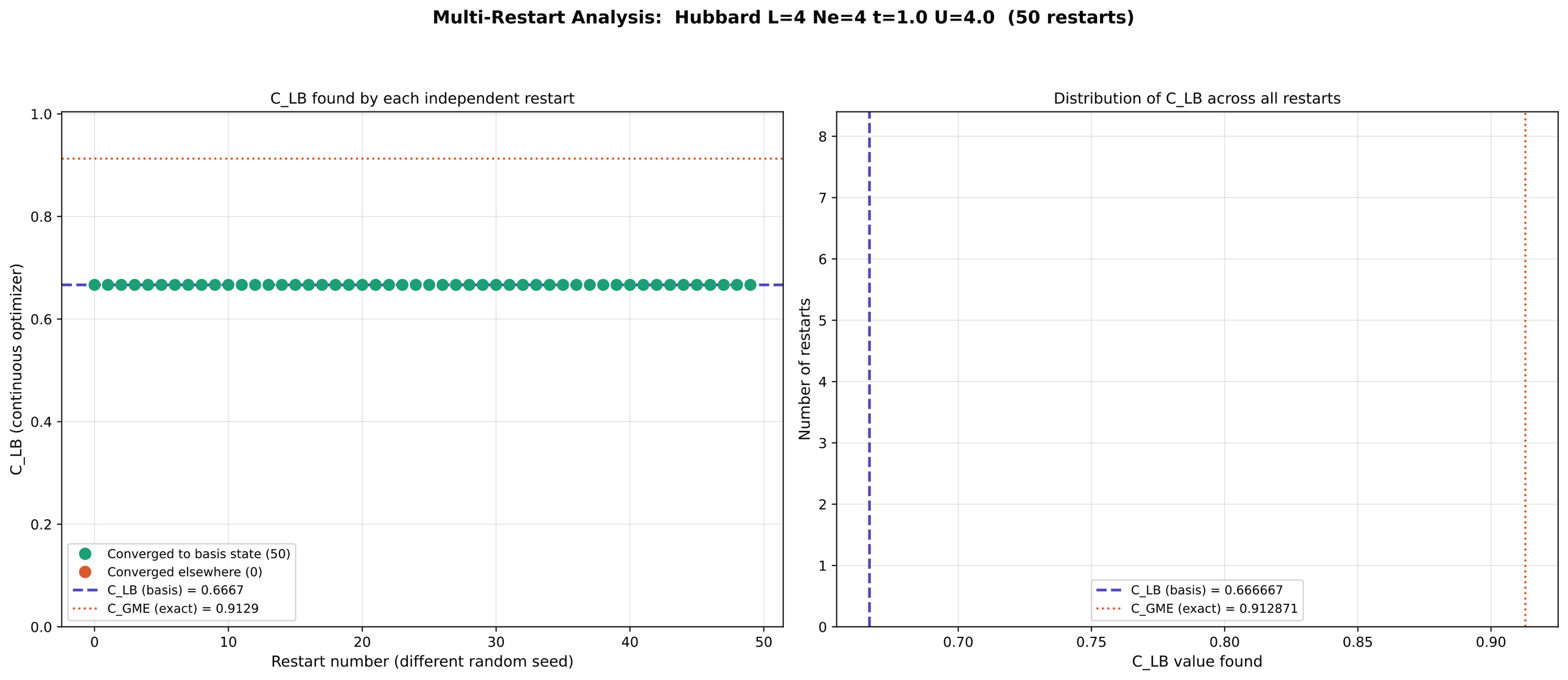
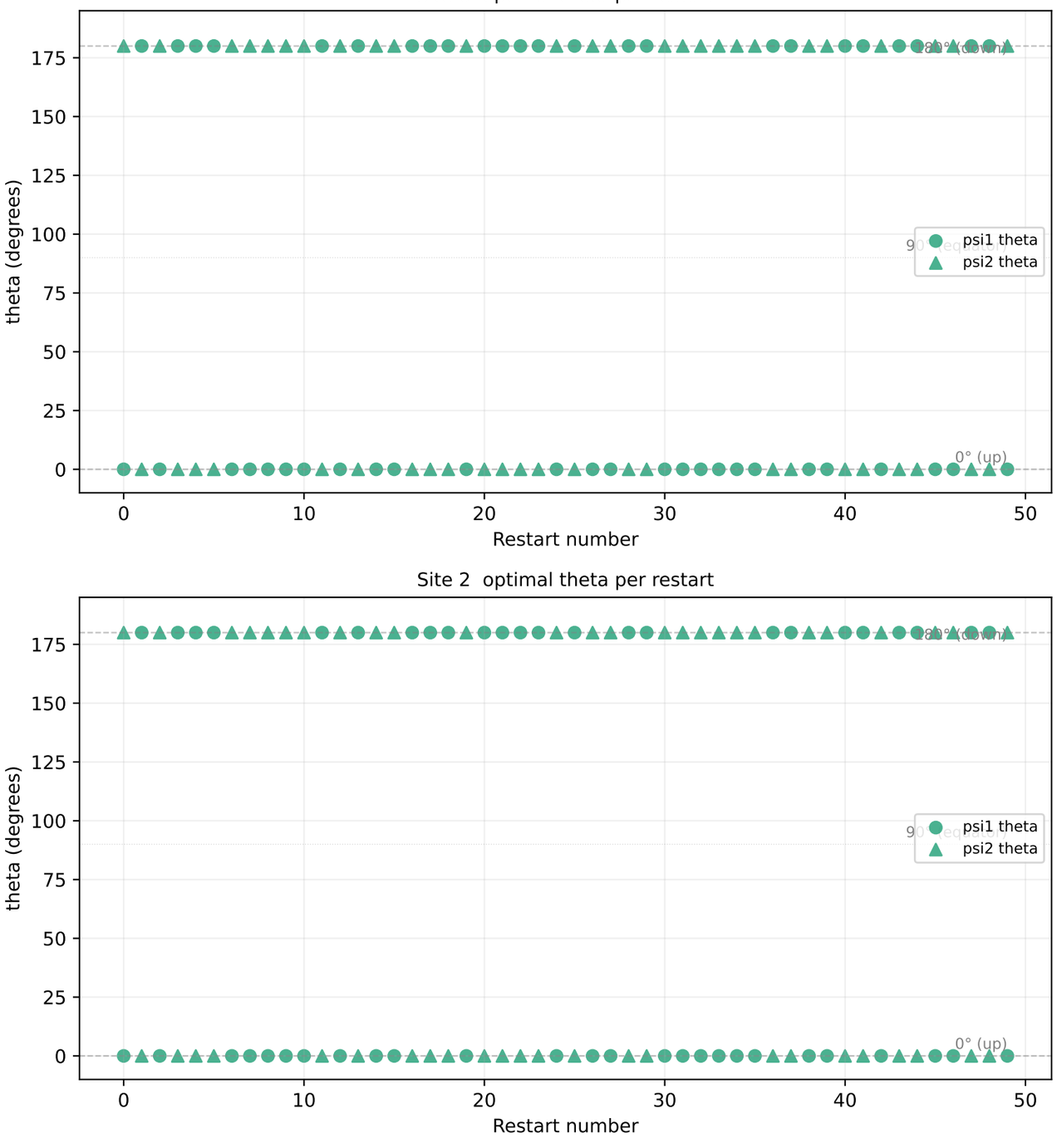
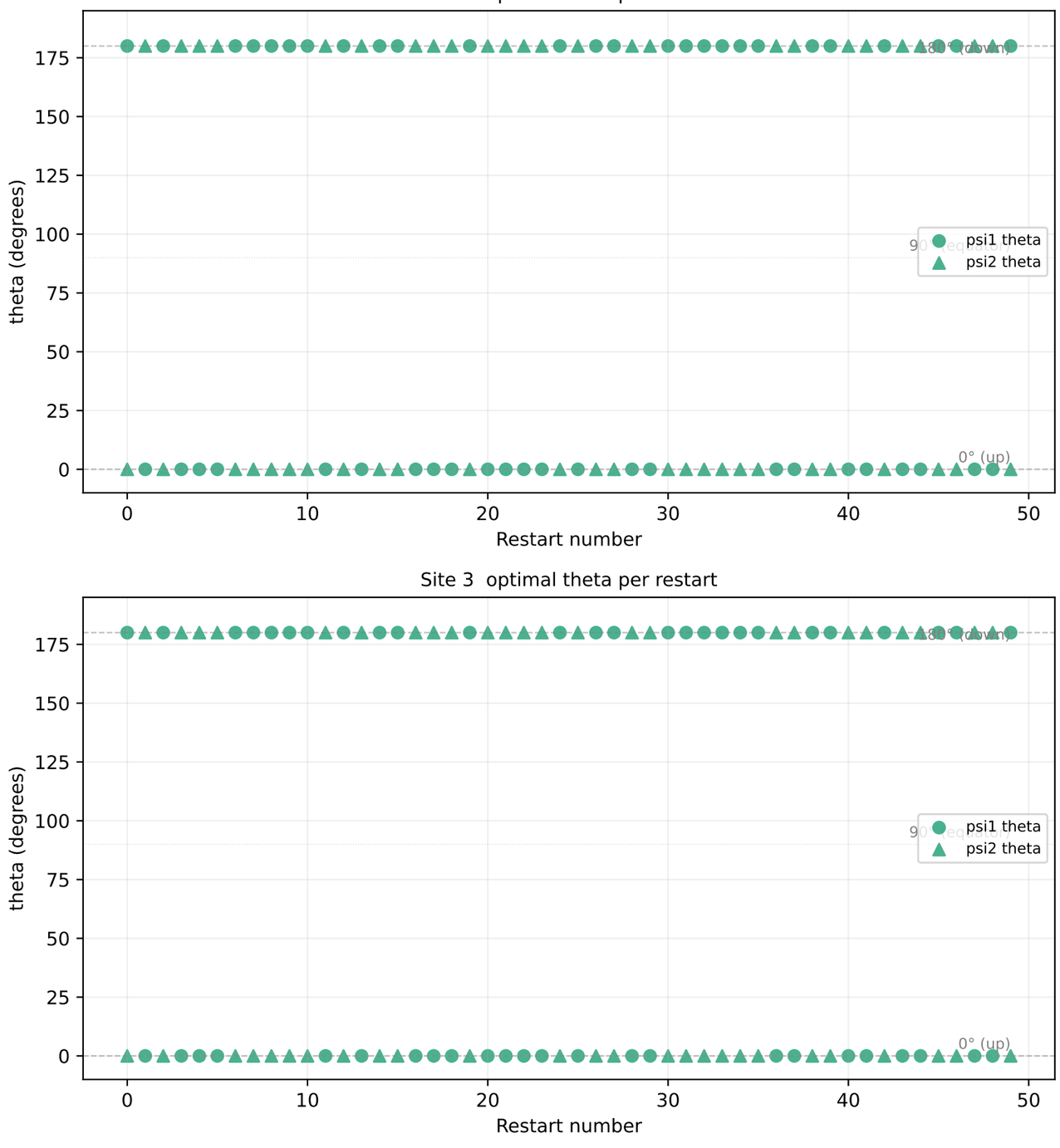

Multi-Restart Analysis: Hubbard L=4 Ne=4 t=1.0 U=4.0 (50 restarts)
C_LB found by each independent restart
C_LB (continuous optimizer)
Restart number (different random seed)
Converged to basis state (50)
Converged elsewhere (0)
C_LB (basis) = 0.6667
C_GME (exact) = 0.9129
Distribution of C_LB across all restarts
Number of restarts
C_LB value found
C_LB (basis) = 0.666667
C_GME (exact) = 0.912871

Optimal theta values found by each restart (degrees)
theta = 0° or 180° corresponds to basis state poles





Multi-Restart summary of Results

```
System:  L=4  Ne=4  t=1.0  U=4.0  mode=uniform
n = N = 4   (pure state RDM,  Tr(rho^2) = 1)

============================================================

  C_GME (exact eq.1)       = 0.91287093
  C_LB  (basis states)     = 0.66666667
  Best C_LB (continuous)   = 0.66666667
  Improvement              = -3.58e-12

============================================================

  Optimizer  : differential_evolution (scipy)
  Parameters : 16  (4 thetas + 4 phis for each of psi1, psi2)
  Restarts   : 50  (each from a different random starting point)

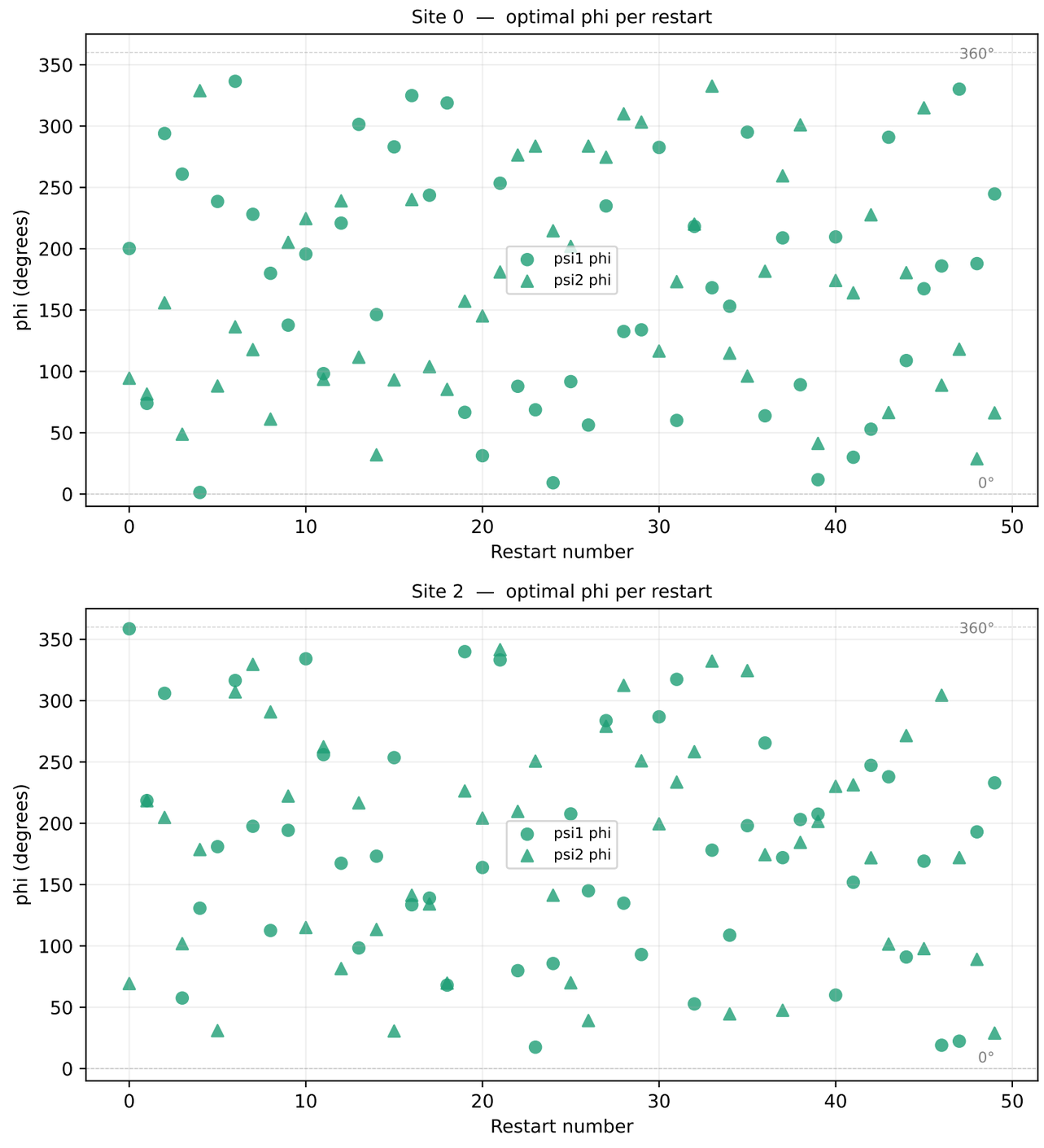
  Restarts converging to basis state : 50 / 50
  Restarts converging elsewhere      : 0 / 50
  Restarts with improvement          : 0 / 50

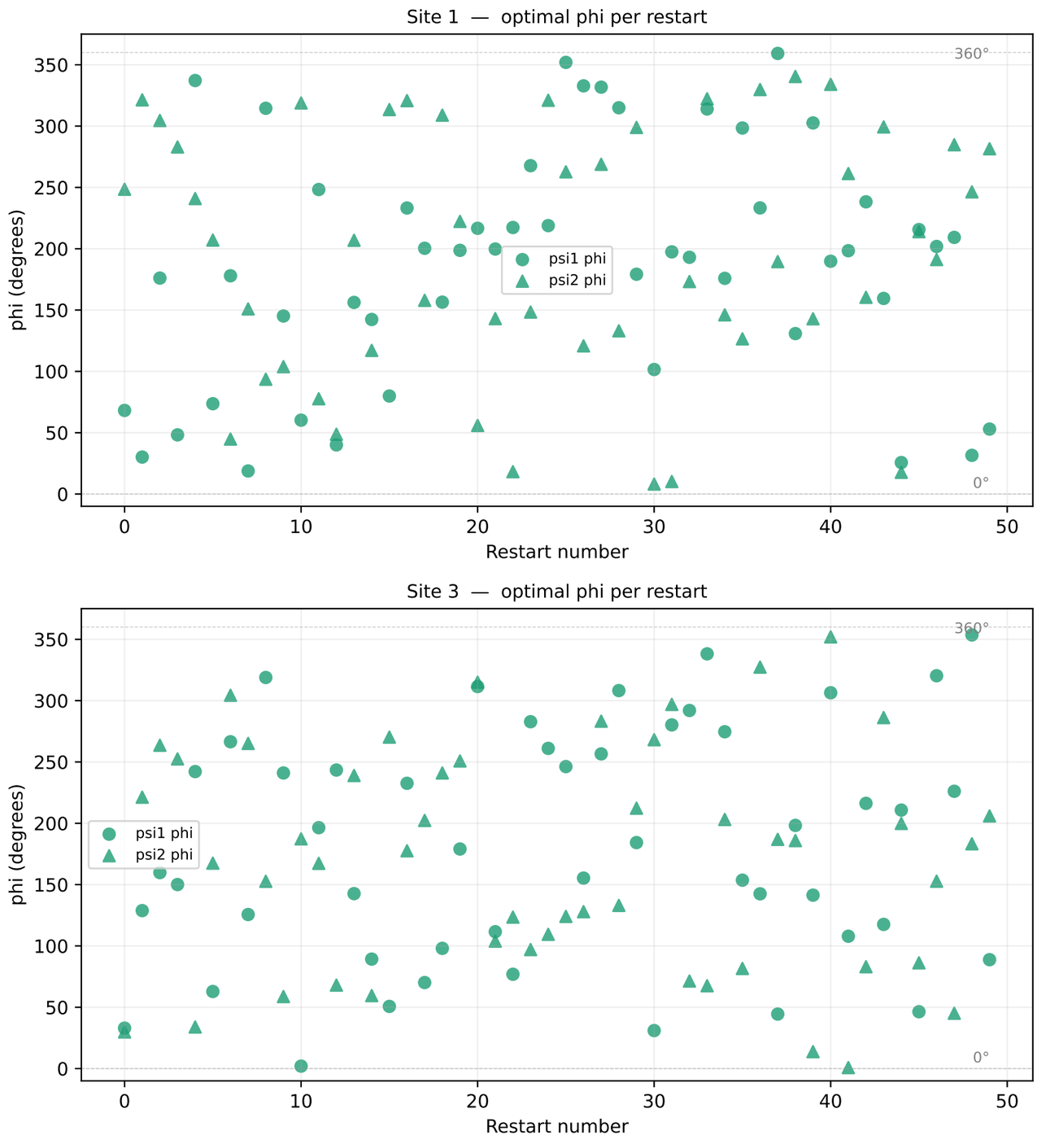
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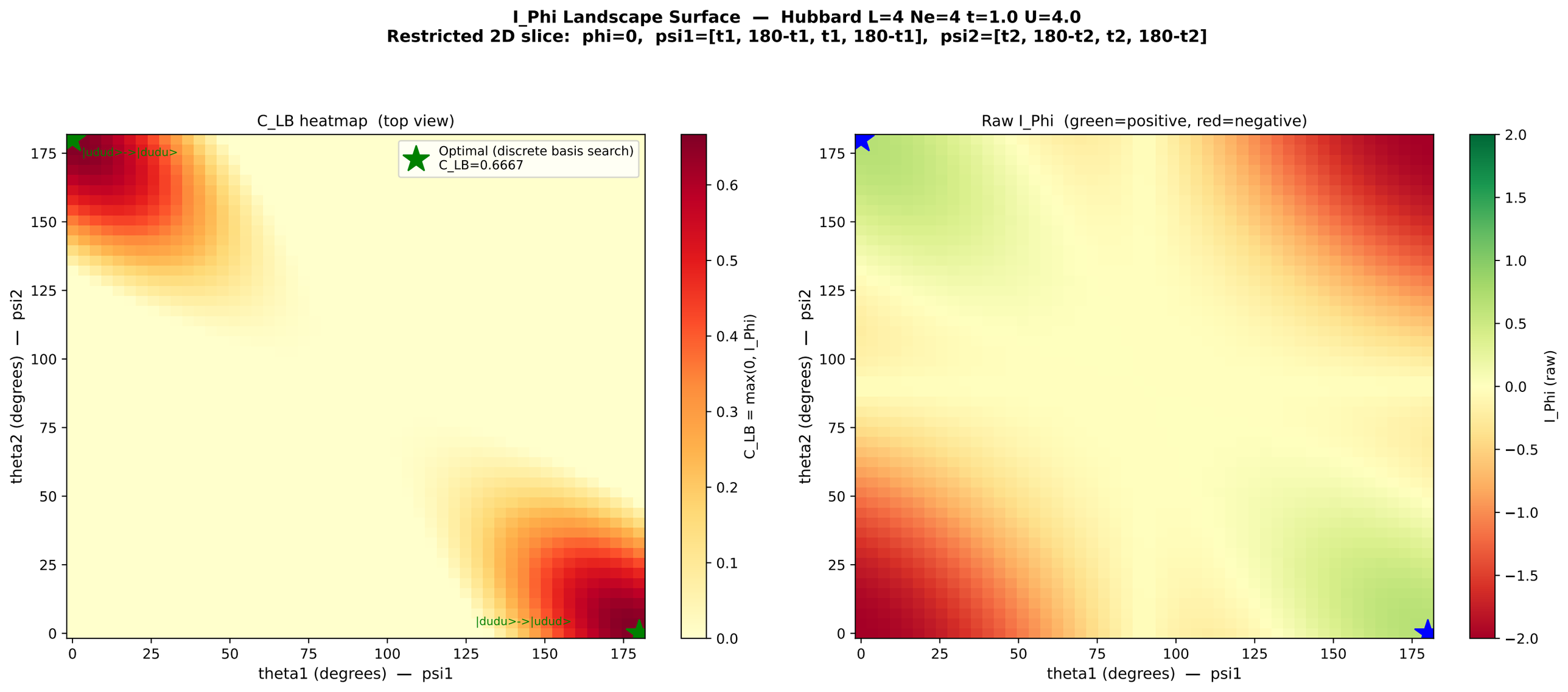
  All 50 restarts converged to the same optimal |Phi>:
  z1 = [0, 1, 0, 1]   z2 = [1, 0, 1, 0]
  (|udud> and |dudu> — the two Neel configurations)

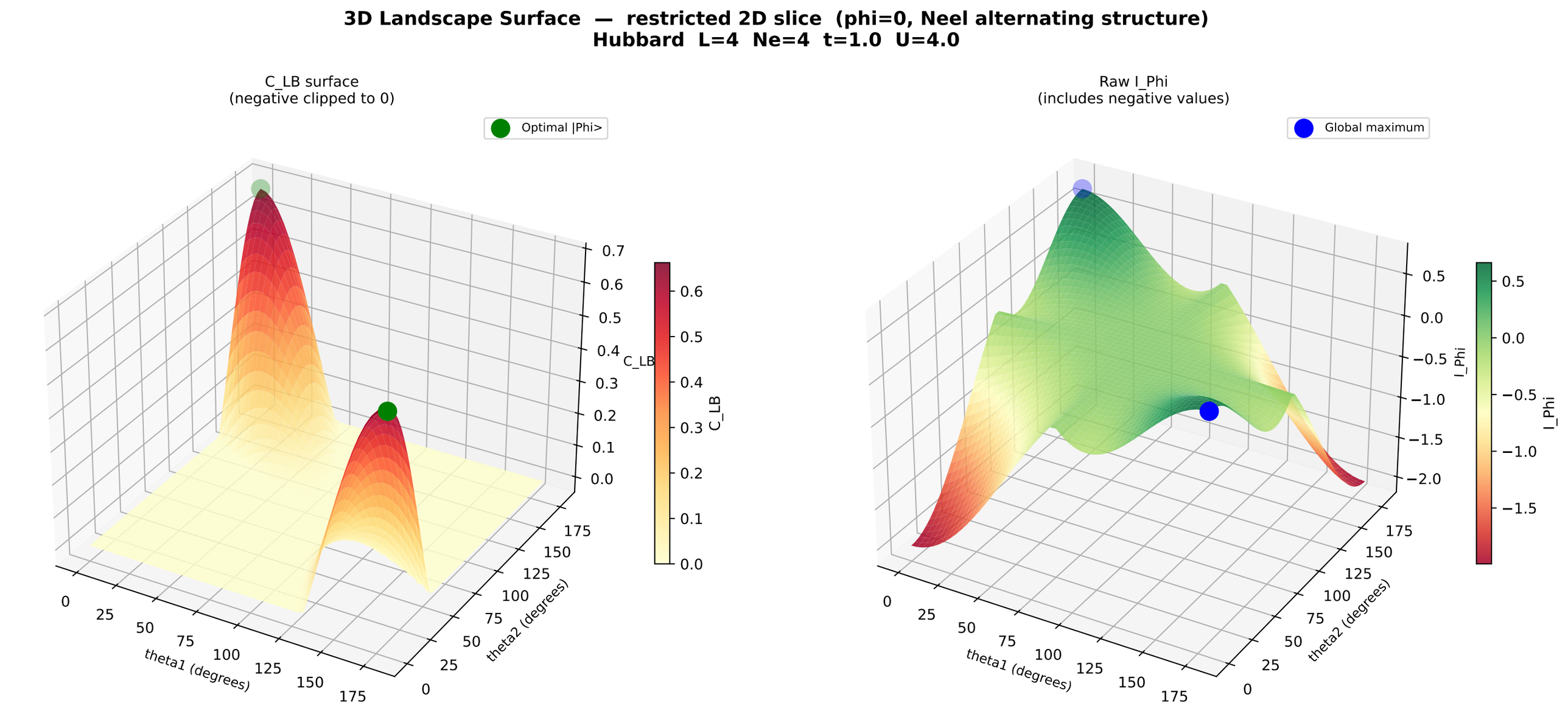
  The phi angles at the optimum are arbitrary
  (phi has no effect when theta = 0 or 180 degrees)
```

Optimal phi values found by each restart (degrees)
phi is arbitrary at the poles (theta = 0° or 180°)






I_Phi Landscape Surface — Hubbard L=4 Ne=4 t=1.0 U=4.0
Restricted 2D slice: phi=0, psi1=[t1, 180-t1, t1, 180-t1], psi2=[t2, 180-t2, t2, 180-t2]
C_LB heatmap (top view)
Optimal (discrete basis search)
C_LB=0.6667
|ududu>->|dudu>
|dudu>->|ududu>
theta2 (degrees) — psi2
theta1 (degrees) — psi1
C_LB = max(0, I_Phi)
Raw I_Phi (green=positive, red=negative)
I_Phi (raw)


3D Landscape Surface — restricted 2D slice (phi=0, Neel alternating structure)
Hubbard L=4 Ne=4 t=1.0 U=4.0
C_LB surface
(negative clipped to 0)
Optimal |Phi>
C_LB
theta1 (degrees)
theta2 (degrees)
Raw I_Phi
(includes negative values)
Global maximum
I_Phi
theta1 (degrees)
theta2 (degrees)